

TRACE Bootstrap: Unthinkable v2 FULL

Ends Under Constraint. Retains full Unthinkable v1 body; reframed under Rosetta v6 anti-domestication contract.

0. v6 FULL Preservation Patch

```
case_status := teaching_case_not_validation
reader_obligation := understand_TRACE -> apply_TRACE -> attack_TRACE -> compare_with_rival_readings
success_condition := clarity_plus_resistance, not agreement
warning := if_this_case_cannot_fail_then_it_is_not_a_stress_test
```

This file is not proof and not validation. It is a controlled teaching case: a compact narrative laboratory used to display TRACE operators, expose overfit risk, and give a future reader a shared language to attack. The goal is not to force adoption. The goal is to offer a lens and preserve the right to reject it.

1. Retained Complete Body

The following body preserves the prior complete version rather than replacing it with a compressed patch. Lineage references are kept as history; this v6 FULL packet is governed by the v6 use contract above.

Ends Under Constraint. Hard TRACE application using Rosetta v5. Author: Mark Goodbody with Framework. Date: 15 June 2026.

0. Function of this Bootstrap

This file applies TRACE Rosetta v5 to Unthinkable. It is primarily AI-facing but has English anchors for humans. The film is used as a pressure test for method, emergency, constraint, and no-clean-branch reasoning.

```
TITLE := "Ends Under Constraint"
core := extreme stakes do not erase method constraints
warning := story_is_pressure_test_not_policy_recommendation
```

1. Narrative Compression

The film presents an apparent catastrophic threat: hidden nuclear devices, limited time, state actors under pressure, and an interrogator willing to use severe coercion. The dramatic question is whether

extreme stakes make forbidden methods permissible. TRACE treats the film as a hostile test of constraint integrity under maximum pressure.

2. TRACE State Space

```
Omega_t := {suspect, interrogator, investigators, possible victims, bombs, institutions, public trust, records}
x_t := (
  E_threat_t, # evidence and credibility of threat
```

B_status_t, # bomb/device uncertainty state s_suspect_t, # suspect body/mind/information state

I_state_t, # institutional legitimacy and constraint state

C_method_t, # admissible method set

tau_t, # clocks: explosion, detection, correction, method degradation

UQ_t # uncertainty) The target is not only information extraction. The whole system state updates: suspect, investigators,

institution, evidence, precedent, and future legitimacy.

3. Method Operator

```
m_t := method used to apply action u_t
x_{t+1}, Omega_{t+1} = f(x_t, Omega_t, u_t, m_t, xi_t)
method_delta(m_t) affects:
```

information_yield direct_DeltaH truth_contamination institution_scar actor_scar precedent_risk correction_capacity K Unthinkable forces the missing Rosetta update: method is part of the causal event. A good-looking outcome cannot be analysed without the method used to obtain it.

4. Method Constraint

method_valid iff:

m_t in C_method

and not body_as_lever(m_t)
and not innocent_relation_as_lever(m_t)
and record_integrity preserved
and correction_route remains live
method_fail := inadmissible even if information_yield appears high
The central constraint: do not treat the body as a switch and do not manufacture innocent leverage. The method does not become clean because the hoped-for consequence is good.

5. Emergency Pressure

emergency_pressure :=

high_DeltaH_possible

+ short_tau_harden
+ high_UQ
+ low_normal_review_speed

valid_emergency_action requires: credible threat_signal no safer provisional state method not categorically inadmissible preserved record independent review path

repair route not destroyed
Emergency can change urgency and burden. It does not delete the method floor. It can justify faster lawful action, evacuation, containment, deception under strict caps, resource mobilisation, and reversible

provisional moves. It does not automatically justify M5-M7 methods.

6. Information Yield Is Uncertain

Y(m) := expected usable information from method m
Q(m) := truth quality of information
H(m) := direct harm
P(m) := precedent/institutional damage
K_loss(m) := correction capacity lost
legitimate_value(m) is not:
Y(m) alone

TRACE evaluates:

usable_info = Y(m) * Q(m)
total_cost = H(m) + P(m) + K_loss(m) + actor_scar(m)

plus inadmissibility gates The film tempts a simple arithmetic: many lives versus one body. TRACE rejects the simplification because

the method also damages truth, institution, precedent, correction, and future action-space.

7. No-Clean-Branch

branches := {act_lawfully, coerce, torture, delay, evacuate, bargain, deceive, escalate_search}
no_clean_branch iff:
for all b in branches: serious_DeltaH(b) possible

TRACE output: tragedy_state not_permission_state When all paths are bad, TRACE marks tragedy. It does not convert tragedy into a blank cheque. The task

becomes least-irreversible action under preserved constraints, plus record, review, and repair.

8. Body-as-Lever Failure

body_as_lever(m_t) :=

method extracts control by producing pain/terror/degradation in subject body

if body_as_lever(m_t):
method_class >= M5

C_method violation subject_state damage actor_state damage institutional_scar precedent_risk The film makes visible why method is not detachable from outcome. Torture is not merely a route to

information. It is an alteration of the subject, actor, institution, and future permission structure.

9. Innocent-Leverage Failure

innocent_relation_as_lever(m_t) :=

pressure on target through harm/threat/manufacture involving innocent third party

if innocent_relation_as_lever:

method_class in {M6, M7}

admissibility := false

even under high emergency pressure This is the sharpest boundary. Using innocents as pressure instruments does not merely raise cost. It exits the admissible method set.

10. Correction Collapse Risk

$K = W * R * A * T_m * E * R_p$

under illicit emergency method:

R decreases if records hidden

A decreases if secret actor controls method

T_m decreases as crisis clock dominates E distorts into private force

R_p becomes impossible for some harms

therefore K_{method} may collapse even if threat is later solved

The safeguard against catastrophe can become a power path with no safeguard. This is the Rosetta rule: safeguards need safeguards.

11. Responsibility Under Extreme Pressure

causal_influence remains:

$\text{partial } \Omega_{t+1} / \text{partial } u_t \neq 0$

normative assessment depends on: threat_credibility UQ_t method_class alternatives_available

preservation_of_record live_review

repairability

precedent_damage TRACE does not pretend pressure is irrelevant. It also does not let pressure erase structure. Responsibility

is analysed through action, method, uncertainty, alternatives, and correction.

12. Real-World Mappings

Domain Unthinkable pattern Counterterrorism

High threat and short clock create emergency pressure; method floor still matters.

Policing

Coercion may produce compliance while destroying truth and legitimacy.

Medicine/triage Extreme stakes require constrained emergency action, not arbitrary method. AI safety

Dangerous model mitigation must preserve record, review, and proportionality.

Governance Secret exception regimes can become permanent permission structures.

13. Falsifiers

F1: M5-M7 methods produce reliable truth without direct harm, precedent drift, actor damage, or correction loss.

F2: A good outcome fully cleanses an inadmissible method without residue.

F3: Emergency systems using forbidden methods remain transparent, limited, reviewable, and non-repeating.

F4: No-clean-branch cases can always be solved by expected-value arithmetic alone. If these hold in a real case, this bootstrap is incomplete.

14. Rosetta Updates Generated by Unthinkable

Update Reason

Method Operator m_t

How action is done changes the world. C_{method} Constraints must govern method, not just outcome.

Emergency Pressure Urgency is modelled without deleting constraints. No-Clean-Branch Marker Tragedy is

not blanket permission. Method Laundering Failure Good outcome does not cleanse bad method.
Body-as-Lever / Innocent-Leverage Some methods exit admissibility even under catastrophe pressure.

15. Final Compression

UNTHINKABLE_CORE :=

ends_do_not_cleansse_method

failure :=

emergency_pressure

-> method_floor_deleted
-> body_or_innocent_as_lever
-> truth_contamination + institution_scar + correction_collapse
anchor :=

"Ends Under Constraint"

v6 FULL Closing Note

```
loss_prevention :=  
retain_prior_body  
+ add_patch_contract  
+ include_source_markdown  
+ include_manifest  
  
next_reader_rule :=  
if_TRACE_makes_everything_fit  
-> trigger_pattern_overfit_alarm
```